

My Product Manager Download API

Contents

Introduction.....	1
Environments.....	1
Security	2
Versioning	4
Filtering	4
Support for filtering on one trade item	5
Support for looping through all trade items.....	5
Pagination	6
FAQ.....	6

Introduction

The purpose of this API is to give data recipients access to product information stored in the 'My product Manager' application of GS1 Belgium & Luxembourg. The product information within this system is created by the information providers (often the brand owners of the products). They define which information is available for the data recipients.

Environments

GS1 provides two environments that can be connected to. All API calls during development can be executed against the **UAT environment**. The hostnames can be switched to the production environment when your application is promoted to production.

UAT:

- Authorization endpoint: <https://login-uat.gs1belu.org/oauth/token>
- API endpoint: [https://api-
uat.gs1belu.org/myproductmanager/download/v2/tradeitems](https://api-uat.gs1belu.org/myproductmanager/download/v2/tradeitems)

PROD:

- Authorization endpoint: <https://login.gs1belu.org/oauth/token>
- API endpoint: <https://api.gs1belu.org/myproductmanager/download/v2/tradeitems>

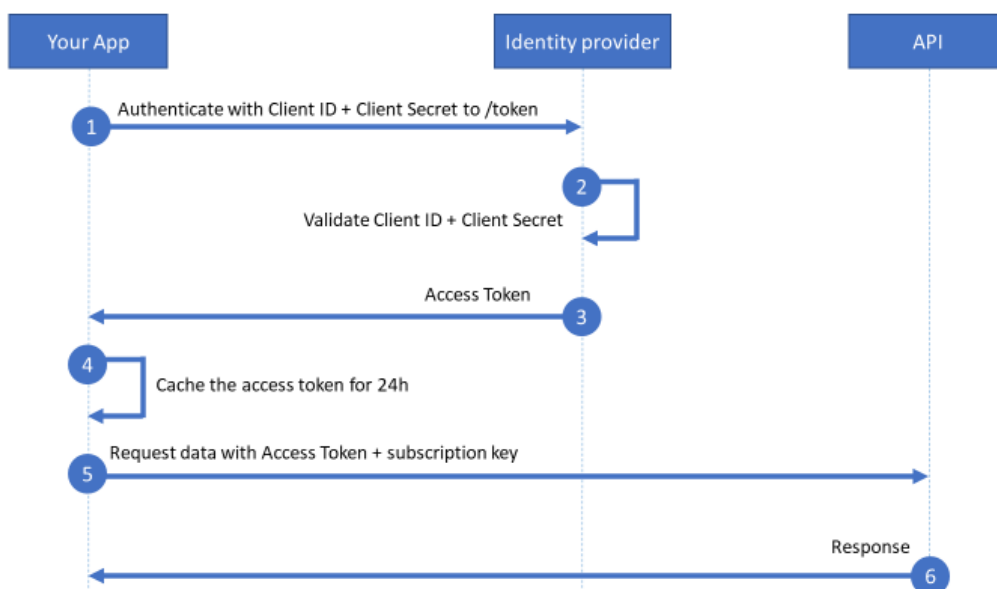
PLEASE NOTE: the API endpoint contains the version of the JSON contract (e.g. V2). The latest contract information is available at:
<https://www.gs1belu.org/nl/my-product-manager>

Security

This API is secured by OAuth 2.0, specifically with the client credentials flow. This flow allows the application on your server to be authenticated and authorized with our identity provider.

The high level flow is as follows:

1. Your app authenticates with the Authorization Server using its **Client ID** and **Client Secret** on the authorization endpoint.
2. The Authorization Server validates the Client ID and Client Secret.
3. The Authorization Server responds with an **Access Token**.
4. The Access Token is valid for 12 hours. The token **must be cached during that period**.
5. Your application can use the Access Token to call the TradeItem API on behalf of itself.
6. The TradeItem API responds with requested data.



An example of the request to fetch the access token is shown below. Cache the value of "access_token" for 12 hours (please note that the example contains PROD urls)

▼ POST https://login.gsibelu.org/oauth/token

► Network

▼ Request Headers

Content-Type: "application/x-www-form-urlencoded"
User-Agent: "PostmanRuntime/7.23.0"
Accept: "*/*"
Cache-Control: "no-cache"
Postman-Token: "e4a01d09-184b-477c-bdac-41ab23e80d3e"
Host: "login.gsibelu.org"
Accept-Encoding: "gzip, deflate, br"
Content-Length: 193
Connection: "keep-alive"

▼ Request Body

client_id: "[REDACTED]"
client_secret: "[REDACTED]"
audience: "https://api.gsibelu.org/"
grant_type: "client_credentials"

► Response Headers

▼ Response Body 

access_token: "[REDACTED]"

scope: "write:tradeitem compose:cic"
expires_in: 86400
token_type: "Bearer"

To retrieve trade items from the API, pass along the "access_token" received from the login request as a Bearer token using the Authorization header. In addition, your subscription key must be included as the value of the "Ocp-Apim-Subscription-Key" header.

```
GET https://api.gs1belu.org/myproductmanager/download/v1/tradeitems
```

Network

Request Headers

Ocp-Apim-Subscription-Key: [REDACTED]

Authorization: Bearer [REDACTED]

access_token

For security reasons we have implemented a limit in the number of calls our users can perform on the system.

- **Download**

- 10 calls/second (600/min window)

- **Upload**

- 10 calls/second (600/min window)
- 10k calls/day

Versioning

This API will possibly change in the future. This API will follow Postel's Law: "Be conservative in what you do, be liberal in what you accept from others." This law has some implications for an API consumer and provider.

As a consumer:

- Ignore new and unknown fields in the payload (cfr. tolerant reader pattern).
- Be prepared for new enumeration values. Provide default behaviour for unknown values, if possible.

As a provider, we will:

- Only add optional fields.
- Never change the meaning or type of a field.

The API will increase the version when the needed changes are no longer possible to implement in the current API without breaking changes. GS1 will try to support the old API version for three months before deprecation. This allows consumers time to start using the new API version.

All version increases will be communicated via mail to the registered mail address. Following RFC8594, the sunset header will be used to indicate the moment in time when a version of the API will not be responsive. Best practice is to implement monitoring on this header, as well as preparing to process dates in the past.

Filtering

The API supports two use cases: Looping through all information for all trade items and querying the API for information of one trade item defined by the GTIN. Both use cases are supported by query parameters for paging and filtering.

Support for filtering on one trade item

The API can be used to return only information about one GTIN.

- **“gtin”**: Get all information about one specific GTIN. If one GTIN is linked to multiple trade items, all will be returned. This is possible when 1 GTIN is uploaded by multiple informationProviders in My Product Manager (for example by the brand owners, a wholesaler, importer, etc.)
 - E.g. <https://api.gs1belu.org/myproductmanager/download/vxx/tradeitems?gtin=05410000999993>
- **“brandOwnerDefault”**: Set the query parameter to true to only get one trade item for the requested GTIN. The API will return all information of the trade item of which the actual brand owner is the information provider. This filter is only relevant when the brandOwnersGLN was filled in by the data supplier.

Support for looping through all trade items

The download set can also be filtered with the following query parameters:

- **“informationProviderGLN”**: By adding this query parameter, you can specify that you would only like to get data from a specific information provider. The GLN is a worldwide unique 13 digit number that identifies the party who is providing the product information.
 - E.g. <https://api.gs1belu.org/myproductmanager/download/vxx/tradeitems?informationProviderGLN=5410000999993>
- **“since”**: By adding this query parameter, you can specify that you would only like to get the trade items that are new or were updated since the indicated timestamp. The API will return only trade items of which the **last modified datetime** is later than the provided timing.
 - E.g. <https://api.gs1belu.org/myproductmanager/download/vxx/tradeitems??since=2025-01-29T14:54:18Z>
- **“isDiscontinued”**: By adding this query parameter, you can specify that you would only like to get the trade items that are or aren't discontinued (true or false). If “IsDiscontinued”=true, the API will return only trade items that are discontinued (no longer manufactured or sold by the supplier). If “IsDiscontinued”=false, the API will return only trade items that are not discontinued (still manufactured or sold by the supplier).
 - e.g. <https://api.gs1belu.org/myproductmanager/download/vxx/tradeitems?isDiscontinued=false>
- **“countriesOfSale”**: By adding this query parameter, you can specify that you would only like to get the trade items that are sold on a specific market (by specifying the countriesOfSale code, e.g. 056 for Belgium).
 - E.g. <https://api.gs1belu.org/myproductmanager/download/vxx/tradeitems?countriesOfSale=056>

Pagination

The response is structured according to the [HAL format](#). The links section contains a link to the current page and the next page. Use these links to paginate through the whole list. The page size can be adjusted by the "limit" query parameter. The maximum for the page size is limited to 100.

An example is added below to show how to loop through all the pages.

```
{
  "count": 10,
  "_links": {
    "self": {
      "href": "https://myproducts.gs1belu.org/myproductmanager/download/v1/tradeitems?limit=10"
    },
    "next": {
      "href": "https://myproducts.gs1belu.org/myproductmanager/download/v1/tradeitems?cursor=5d1e3dba81020c1bb4e47c87&limit=10"
    }
  },
  "_embedded": {
    "tradeitems": [
      {
        "gtin": "05410000000064",
        ...
      },
      {
        "gtin": "05407007400003",
        ...
      },
      ...
    ]
  }
}
```

The JSON above is a response of the API. The response contains 10 TradeItems under the "_embedded" key. The links to the current and next page can be found under the "_links" key. Use the URL provided in the "next" link to get the next page.

FAQ

1. Q: Why do I get the error {"error":"access_denied","error_description":"Non-global clients are not allowed access to APIv1"} when requesting a token?

A: Don't forget to add the audience in the request:

```
"audience": "https://api-uat.gs1belu.org/",
```

The "/" at the end should be present!